

Restaurant Data Analysis Report

1. Dataset Overview

- **Description of the Dataset**

The dataset contains information about restaurants, including their location, cuisines, ratings, pricing, online delivery availability, table booking facilities, and customer votes. It is used to analyse restaurant trends, customer preferences, and factors affecting ratings.

```
print(data.columns)
['Unnamed: 0', 'Restaurant ID', 'Restaurant Name', 'Country Code',
 'City', 'Address', 'Locality', 'Locality Verbose', 'Longitude',
 'Latitude', 'Cuisines', 'Average Cost for two', 'Currency',
 'Has Table booking', 'Has Online delivery', 'Is delivering now',
 'Switch to order menu', 'Price range', 'Aggregate rating',
 'Rating color', 'Rating text', 'Votes']
```

- **Number of rows and columns**

```
print(data.shape)

(9552, 22)
Rows - 9552
Columns - 22
```

2. Data Cleaning

- **Missing values identified**

```
print(data.isnull().sum())

Unnamed: 0      0
Restaurant ID    0
Restaurant Name  0
Country Code     0
City            0
Address         0
Locality        1
Locality Verbose 1
Longitude       1
Latitude        1
Cuisines       10
Average Cost for two 1
Currency        1
```

```

Has Table booking    1
Has Online delivery  1
Is delivering now     1
Switch to order menu 1
Price range          2
Aggregate rating      2
Rating color         2
Rating text          2
Votes                2

```

- **Data type changes performed**

1. Verified data types of all columns.

```
print(data.dtypes)
```

```

Unnamed: 0          object
Restaurant ID       object
Restaurant Name     object
Country Code       float64
City               object
Address            object
Locality           object
Locality Verbose   object
Longitude          object
Latitude           object
Cuisines            object
Average Cost for two  object
Currency           object
Has Table booking   object
Has Online delivery object
Is delivering now    object
Switch to order menu object
Price range         float64
Aggregate rating     float64
Rating color        object
Rating text         object
Votes               float64

```

2. Converted data types where necessary

```

print(data['Average Cost for two'].dtype)
data['Average Cost for two'] = pd.to_numeric(data['Average Cost for
two'],errors='coerce')
print(data['Average Cost for two'].dtype)

```

```

object
float64

```

- **Duplicate records (if any)**

```
print(data.duplicated().sum())
```

```
0
```

3. Data Analysis

- **Statistical Summary**

```
print(data.describe)
```

```
print(data['Average Cost for two'].mean())
```

```
print(data['Average Cost for two'].median())
```

```
print(data['Average Cost for two'].min())
```

```
print(data['Average Cost for two'].max())
```

```
1199.326387434555
```

```
400.0
```

```
0.0
```

```
800000.0
```

- **Analysis Result**

```
#a. How many unique cities are present in the dataset?
```

```
print(data['City'].nunique())
```

```
142
```

```
#b. Which city has the highest number of restaurants?
```

```
print(data['City'].value_counts().idxmax())
```

```
New Delhi
```

```
#c. Which cuisine appears most frequently?
```

```
data['Cuisines'].dropna().str.split(',').explode().value_counts().idxmax()
```

```
'North Indian'
```

```
#d. Find the average restaurant rating.
```

```
print(data['Aggregate rating'].mean())
```

```
2.6663141361256546
```

```
#e. Identify the restaurant with the highest number of votes.
```

```
print(data.loc[data['Votes'].idxmax(), 'Restaurant Name'])
```

```
Toit
```

```
#f. Find the average cost for two across all restaurants.
```

```
#print(data['Average Cost for two'].dtype)
```

```
#data['Average Cost for two'] = pd.to_numeric(data['Average Cost for
two'],errors='coerce')
print(data['Average Cost for two'].mean())
1199.326387434555
```

```
#g. Compare the average ratings of restaurants that:
#a. Have Online Delivery
#b. Do Not Have Online Delivery
print(data.groupby('Has Online delivery')['Aggregate rating'].mean())
```

```
Has Online delivery
No    2.465192
Orange NaN
Yes   3.248837
```

```
#h. Compare the average ratings of restaurants that:
#a. Have Table Booking
#b. Do Not Have Table Booking
print(data.groupby('Has Table booking')['Aggregate rating'].mean())
```

```
Has Table booking
3.2    NaN
No    2.559283
Yes   3.441969
```

```
#i. Identify the top 10 cities with the highest number of restaurants.
print(data['City'].value_counts().head(10))
```

```
City
New Delhi    5473
Gurgaon      1118
Noida        1080
Faridabad    251
Ghaziabad    25
Ahmedabad    21
Bhubaneswar  21
Lucknow      21
Guwahati     21
Amritsar     21
Name: count, dtype: int64
```

- **Key Observations**

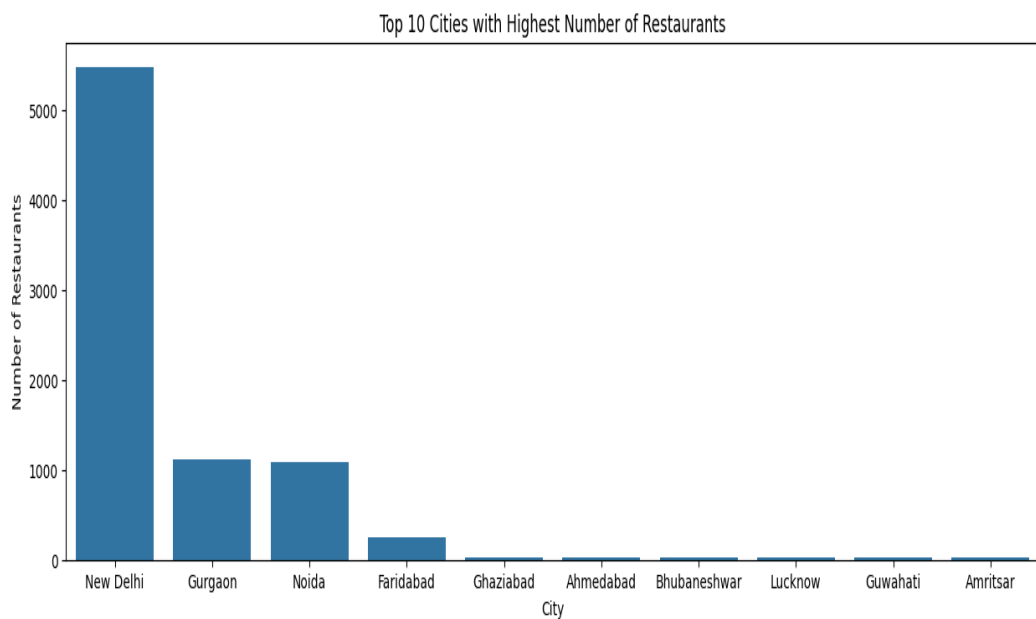
- Certain cuisines are significantly more popular than others.

- *Higher-rated restaurants generally receive more customer votes.*
- *Online delivery and table booking services tend to influence customer ratings positively.*

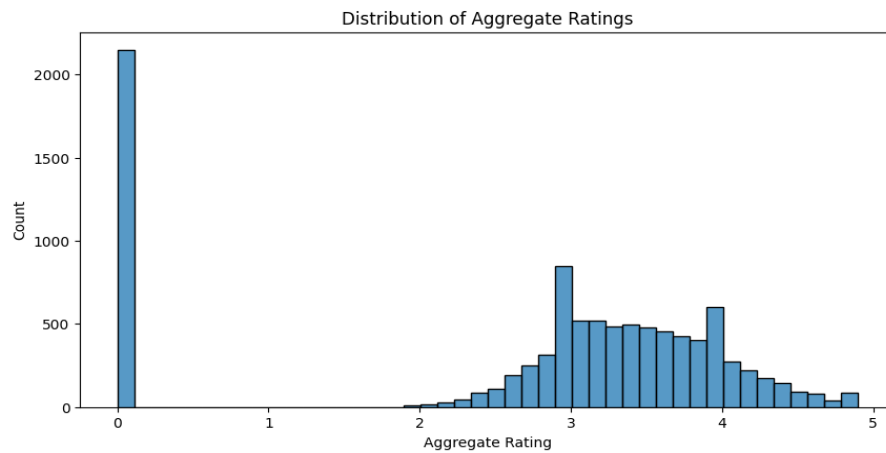
4. Data Visualization

- **Top 10 Cities by Number of Restaurants**
- **Observation:** Shows which cities have the highest restaurant concentration.

City	
New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Ahmedabad	21
Bhubaneswar	21
Lucknow	21
Guwahati	21
Amritsar	21



- **Distribution of Aggregate Ratings**
- **Observation:** Displays the spread of customer ratings and identifies common rating ranges.



- **Restaurants Offering Online Delivery**
- **Observation:** Compares restaurants that provide online delivery versus those that do not.

Has Online delivery

No 7099

Yes 2451



- **Top 10 Most Common Cuisines**
- **Observation:** Identifies the cuisines most frequently offered by restaurants.

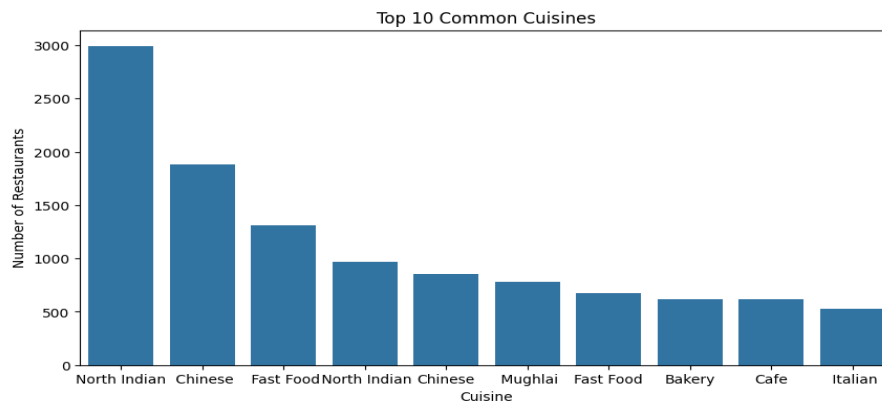
Cuisines

North Indian 2992

Chinese 1880

Fast Food 1314

North Indian 968
Chinese 855
Mughlai 780
Fast Food 672
Bakery 621
Cafe 617
Italian 530



- **Average Rating by Price Range**
- **Observation:** Shows the relationship between restaurant pricing and customer ratings.

Price range
 1.0 1.999887
 2.0 2.941054
 3.0 3.683381
 4.0 3.818974



5. Conclusion

- **Final Insights**

- *Restaurant ratings are influenced by service features such as online delivery and table booking.*
- *Popular cuisines attract a large customer base and dominate the dataset.*
- *Higher-priced restaurants often maintain competitive ratings due to service quality and customer experience.*
- *Customer votes and ratings are valuable indicators of restaurant popularity.*

- **Business Observations**

- *Restaurant owners can improve customer satisfaction by offering online delivery and table booking services.*
- *Popular cuisines can be prioritized when planning menu expansions.*
- *Businesses can use location trends to identify promising markets for new restaurant openings.*